Quality Management

8 — Digital images: pixels, grey levels, colour and files

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Today

From an image of light to a matrix of numbers

Lecture 7 ended with an image of light on the sensor. Today it becomes **numbers** — and numbers are what make an image something you can measure, compare, store and decide on.

0 **Recap and why**
lecture 7 in two slides; why digital

1 **Pixels**
sampling, aliasing, resolution

2 **Grey levels**
quantisation, bit depth, saturation

3 **Colour**
channels, colour spaces, measuring colour

4 **Files and records**
formats, compression, images as evidence

From lecture 7: forming the image

Recap, 1 of 2

Light and colour

Visible light: 380–700 nm. An object's colour depends on the object and on the light. The eye has three kinds of cone, so three channels are enough to reproduce a colour.

Image formation

Pinhole and lens: similar triangles. $1/f = 1/u + 1/v$, magnification $m = v/u$. Field of view $W = S(u - f)/f$.

Errors of the optics

Distortion, vignetting, chromatic aberration, perspective. Corrected by calibration on known targets, or avoided with telecentric lenses.

Where we are in the chain

light from the part → lens → **an image of light on the sensor** → *numbers* → decision. Everything before the arrow in bold was lecture 7; the step after it is today.

From lecture 7: the settings and their price

Recap, 2 of 2

Setting	Gives	Costs
Focal length & distance	the field of view; less perspective from far	detail per part of the field
Aperture (larger *N*)	more depth of field	light (half per stop); diffraction
Exposure time (shorter)	no motion blur	light
Gain (ISO)	brightness	noise; earlier saturation

The rule

Decide from what must be seen; choose the smallest *N* that gives the depth; make up the light with lighting, not gain.

What the optics leave out is gone

Blur, saturation and missing depth cannot be repaired by any processing. The digital step that follows can only keep — or lose — what the optics delivered.

Digital images are everywhere

Most of them are never looked at by a person: they are measured by software

Phones
Several cameras each; faces unlock the phone; documents scanned into text.

Cars
Cameras keep the lane, read speed signs and brake for pedestrians.

Medicine
Digital X-ray, CT, endoscopy, microscopy of tissue samples.

Shops and logistics
Barcode scanners at the till; parcels sorted by reading their labels in flight.

Traffic and security
Number-plate recognition for tolls and access; speed cameras.

Sport
Goal-line technology and line calls from several synchronised cameras.

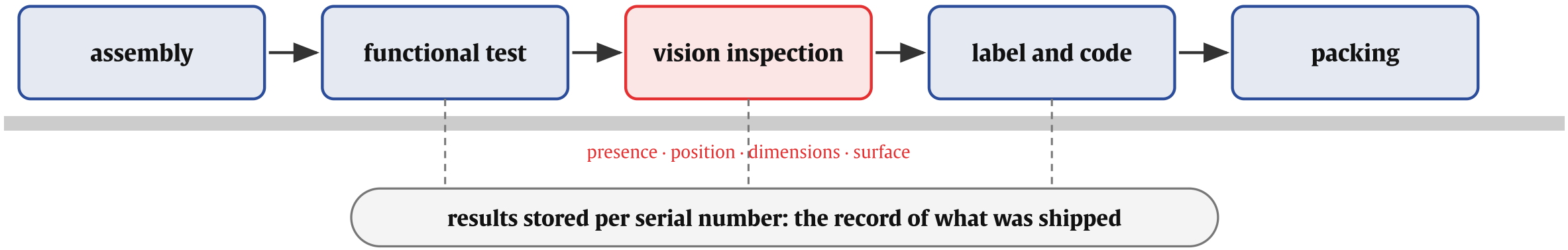
Science
Telescopes, microscopes, satellites: images measured, not just viewed.

Industry
On almost every production line — next two slides.

What these have in common: the image is turned into a **number** or a **decision** — a position, a code, a speed, pass or fail — automatically, many times a second.

In quality management: end-of-line control

The last check before a product leaves the factory



- **End-of-line (EOL) control**: the stations at the end of an assembly line where **every unit** is tested before it is packed — functional tests (does it work?) and inspections (is it complete, correct, undamaged?).
- Cameras do the inspections: presence and position of components, dimensions, surface, labels and codes — in the seconds the line allows for each unit.
- The result of every check is stored with the unit's **serial number**: the record of what was shipped.

The e-scooter of lecture 6

Its end-of-line test was a test ride on a roller bench. A vision station next to it would check that the brake cables are routed, the screws are present, the frame is unscratched and the label matches the model — every scooter, not a sample.

What cameras check, industry by industry

The same few tasks, everywhere

Industry	Typical vision checks
Automotive	gap and flush between body panels · paint defects in an inspection tunnel · clips, connectors and screws present · labels and identification numbers
Electronics	automated optical inspection of circuit boards: every component present, the right way round, every solder joint formed
Pharmaceuticals	blister packs full and tablets unbroken · label and leaflet text verified · a unique code on every pack, read and recorded for traceability
Food and drink	fill level, cap and seal · foreign bodies · sorting fruit, grain and nuts by colour and shape
Metals, plastics, paper	surface defects on strip, film and sheet at full line speed · dimensions of machined and moulded parts
Packaging and logistics	print quality and position · seal integrity · codes read for every case and pallet

Five tasks behind all of them

presence · **position** · **dimension** · **surface** · **identification** (codes and text)

What makes them possible

Every one of these checks is a calculation on the numbers of a digital image — none of them could be done on film.

Why digital: analog and digital images compared

The same optics; a different kind of result

	Analog (film)	Digital
Result	a picture, to be looked at	**numbers**, to be computed with
Available	after chemical development: minutes to hours	at once: hundreds of images per second
Judged by	a person	an algorithm: the same answer for the same image
Inspection	samples	**every part**, at line speed
Response to light	non-linear, depends on the film and its development	linear and calibratable (lecture 9)
Copies	each copy loses quality	identical, as many as needed
Record	prints in an archive	files with time, settings and serial number attached
Cost per image	film and chemicals	almost nothing

Film still has its uses — very large formats, archival stability — but nothing in quality control needs it. The decisive difference is the first row: a digital image is **data**.

A digital image is a measurement

Why measurement science cares

- A 5-megapixel camera makes **5 million measurements at once**, one per pixel, in a fraction of a millisecond.
- Like any instrument it can be **calibrated** — pixels to millimetres, grey levels to light — and its **uncertainty** can be stated.
- Its numbers feed statistics directly: the diameter of every part, straight into a control chart (lecture 6).

Full-field methods

Methods that measure a whole field at once exist only because images are numbers: **digital image correlation** (displacement and strain on a loaded part), **particle image velocimetry** (velocity of a flow), **thermography** (temperature of a surface), **photogrammetry** (3D coordinates).

What the numbers must keep

For a measurement, the digital image must keep what the optics delivered: enough **pixels** for the detail, enough **grey levels** for the contrast, the right **colour** information, and a **file** that does not change the values. Those are the four parts of today.

PART 1

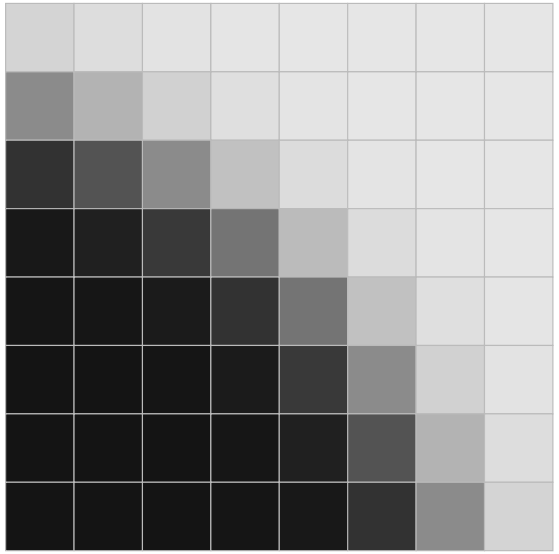
Pixels

The image as a matrix · Sampling · Aliasing · Resolution · Better than a pixel

A digital image is a matrix of numbers

What the camera stores is not a picture

what you see (zoomed in)



what is stored

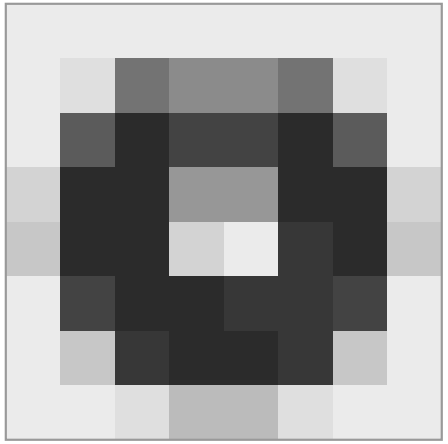
212	221	227	229	230	230	230	230
139	179	209	223	228	230	230	230
50	83	139	193	220	228	230	230
24	32	57	116	187	220	228	230
21	22	27	50	116	193	223	229
20	20	21	27	57	139	209	227
20	20	20	22	32	83	179	221
20	20	20	21	24	50	139	212

the edge of a dark hole in a bright part: about 20 inside, 230 outside

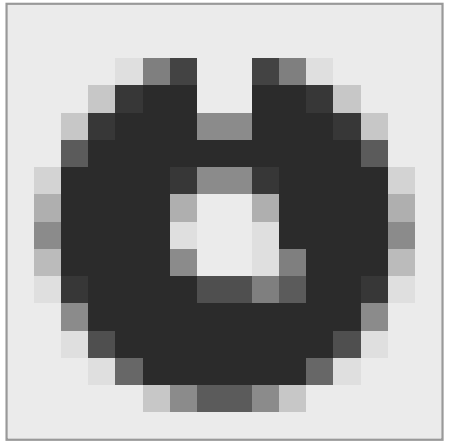
- The light on the sensor is continuous. The sensor is a grid of small light-sensitive cells — **pixels** (picture elements). Each collects the light falling on its area during the exposure and gives **one number**: its grey level.
- Two steps make an image digital: **sampling** — where, the grid of pixels — and **quantisation** — how much, in whole grey levels.
- A grey image is one matrix of M rows × N columns. Pixel (0, 0) is the top-left corner: row first, then column.
- Every inspection algorithm works on these numbers, never on the picture on the screen.

Sampling: how many pixels

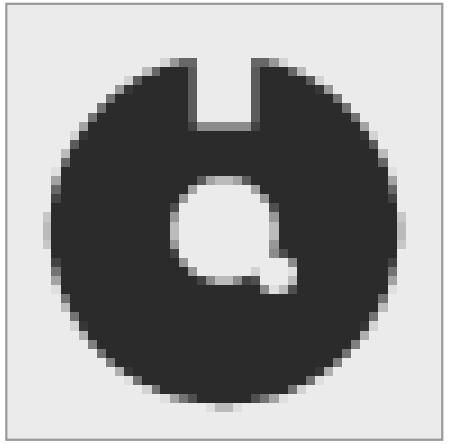
The same part, recorded with three grids



8 × 8 pixels



16 × 16 pixels



48 × 48 pixels

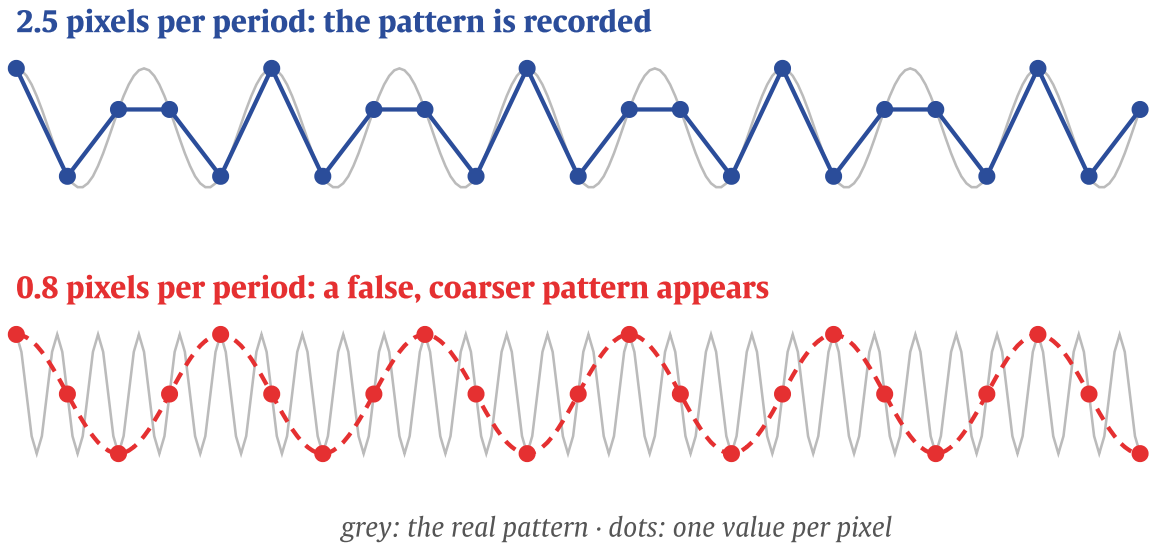
- Each pixel records the **average** light over its area. Detail smaller than a pixel is averaged away: the small hole becomes a grey smudge, the keyway's corners round off.
- The pixel size that matters is the one **on the part**: field of view divided by the number of pixels across it. 100 mm over 2448 pixels is 41 µm per pixel.

More pixels are not free

Twice the pixels across means four times the data, smaller pixels with less light each (lecture 9), and slower processing. Choose the grid from the smallest detail the task needs (slide 14), not from the largest sensor on sale.

Aliasing: when the grid is too coarse

Fine detail does not disappear — it comes back as something false



grey: the real pattern · dots: one value per pixel

- **Sampling theorem** (Nyquist–Shannon): a repeating pattern is recorded correctly only with at least **two pixels per period**.
- A finer pattern is not simply lost: the samples describe a **coarser false pattern** — aliasing.
- You have seen it: **moiré** bands when filming a screen, a striped shirt shimmering on television, wheels that seem to turn backwards in films.
- In inspection, aliasing on a woven fabric, a mesh or machined grooves can look like a defect — or hide one.

Resolution: one word, five meanings

And how many pixels a task really needs

"Resolution" as…	Measured in	Example
pixel count	pixels, megapixels	2448 × 2048 = 5 MP
dots per inch	dpi (scanners, printers)	300 dpi = one dot every 85 µm
pixel on the object	µm per pixel	41 µm on a 100 mm field
optical resolution	smallest detail the lens separates	set by focus, aberrations, diffraction
measurement resolution	smallest change the result shows	can be finer than a pixel (next slide)

Task	Rule of thumb
find a defect	at least 2–3 pixels across it
read printed characters	about 20 pixels of character height
read a 2D code	at least 2–3 pixels per module, more is safer
measure an edge	a fraction of a pixel, with calibration

Rules of thumb from practice; the software's datasheet gives the exact figures.

Better than a pixel

How a measurement can be finer than the grid it is made on

- An edge is not a jump from black to white: it spreads over a few pixels with intermediate grey levels. Where the grey level crosses half-way can be found to a **fraction of a pixel**.
- A feature with a long edge gives many such points. A circle fitted through all of them **averages their random errors**.

A HOLE OF 10 MM

At 50 µm per pixel its edge crosses about **630 pixels** ($\pi \times 10$ mm / 50 µm). Its diameter, from all of them, is far finer than 50 µm.

What averaging cannot fix

Systematic errors — distortion, perspective, uneven light, the choice of grey level — are the same on every point and do not average out. They are removed by **calibration** on a part of known size (lecture 12).

So

The pixel size is not the uncertainty of a measurement, and a smaller pixel does not by itself make a better gauge.

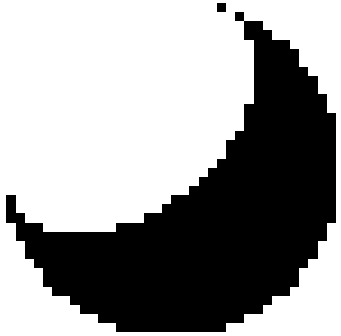
PART 2

Grey levels

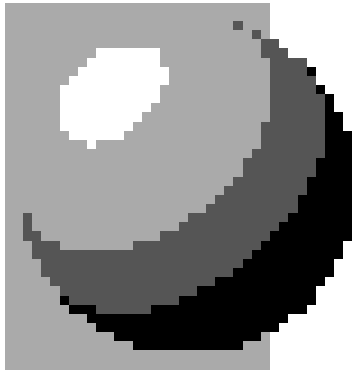
Quantisation · Bit depth · Rounding error · Saturation

Quantisation: how many grey levels

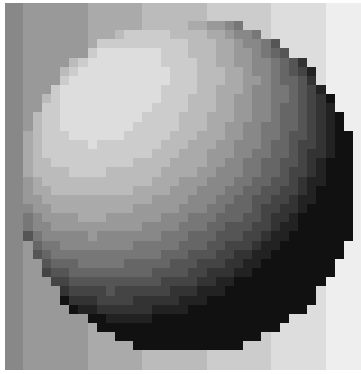
The same image with 2, 4, 16 and 256 levels



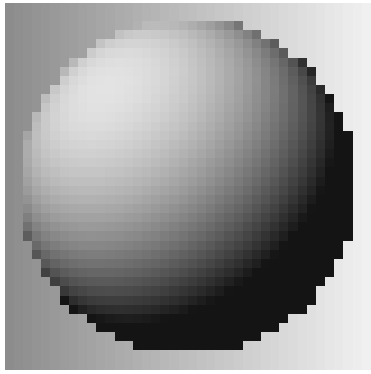
1 bit · 2 levels



2 bits · 4 levels



4 bits · 16 levels



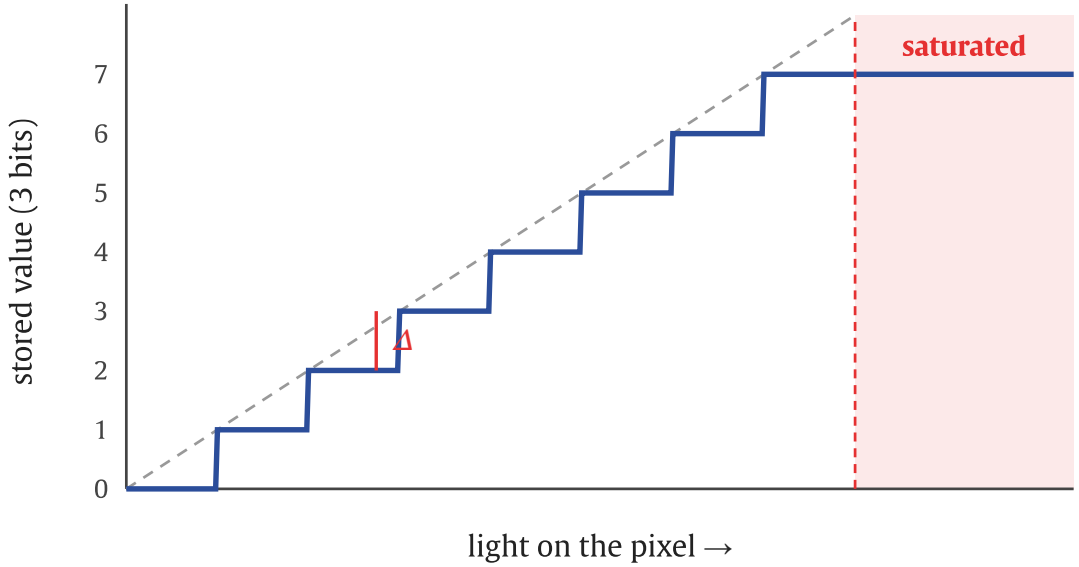
8 bits · 256 levels

- Each pixel's light is rounded to a whole number. With b bits there are $\mathbf{2^b}$ **levels**.
- Too few levels and smooth shading turns into bands (*posterisation*), and small differences of brightness — a faint stain, a shallow dent — vanish into one level.
- 1 bit: black or white. That is what a threshold produces on purpose (lecture 11).

Bits	Levels	Typical use
8	256	screens, most images
10–12	1024–4096	machine vision
12–14	4096–16 384	camera raw files
16	65 536	scientific and medical

Rounding error and saturation

What quantisation adds to every measurement, and what it destroys at the top



- **Rounding error:** between $-\Delta/2$ and $+\Delta/2$ for a step Δ; spread evenly, its standard deviation is $\Delta / \sqrt{12}$ — the same rectangular distribution used for a caliper's resolution in an uncertainty budget.
- **Saturation:** every light above the top level is stored as the top level. A saturated region has no information left — and the edges next to it move.
- **Which bit depth?** Enough that the noise, not the rounding, sets the limit. A high-contrast image — black part on a white backlight — is fine with 8 bits; a faint defect on a bright, shiny surface may need 12.

Check it on every set-up

The histogram of the image (lecture 11) shows at once whether the levels are used well and whether anything sits at the top value.

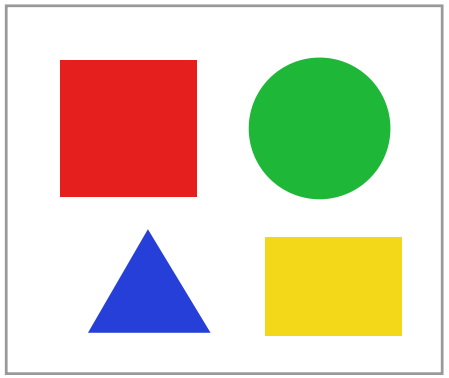
PART 3

Colour

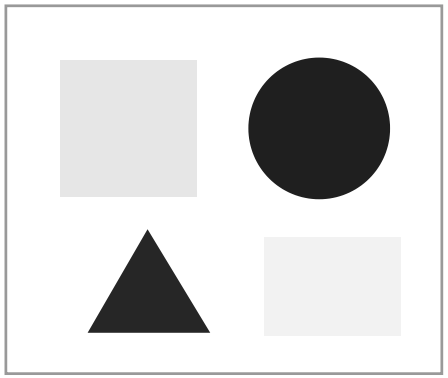
Three channels · Colour spaces · Measuring colour

A colour image is three grey images

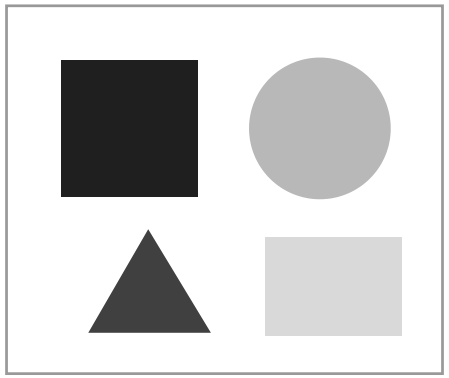
Built to imitate the eye's three kinds of cone (lecture 7)



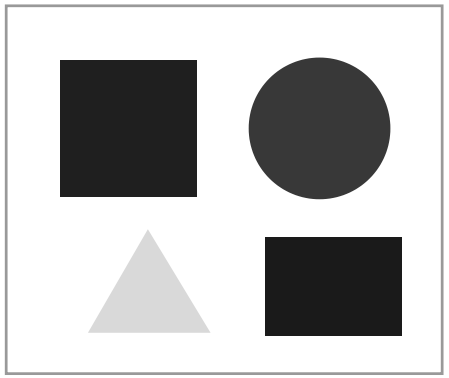
colour image



red channel



green channel



blue channel

- We see colour through three cone types; three measured values per point are enough to reproduce it. So a colour camera records three images — how much **red**, **green** and **blue** light reached each point.
- Each channel is an ordinary grey image. A colour image is three matrices, and each pixel a vector of three numbers: with 8 bits per channel, $2^{24} \approx 16.7$ million colours.

Why it matters in quality management

Colour is a characteristic to inspect — paint shade, ripeness, the colour of a wire in a harness, the colour of an LED — and a tool for inspecting: a defect invisible in grey can stand out in one channel. Choosing the channel where part and background differ most is often the first step of an inspection.

Colour spaces

The same colour, written in the coordinates that suit the job

Space	Coordinates	Used for
RGB	red, green, blue	cameras and screens: how colour is recorded
HSV	hue, saturation, value	colour checks under changing brightness: the hue barely moves when the light dims
CIELAB	lightness L*, a* (green–red), b* (blue–yellow)	**colour tolerances**: distances match what people see; its axes follow the eye's opponent pairs
CMYK	cyan, magenta, yellow, black	printing (lecture 7)
YUV	brightness + two colour signals	video and compression

Why change coordinates? RGB mixes brightness and colour: the same red part, under a dimmer lamp, has completely different RGB numbers. A space that separates them lets you check the colour while ignoring the light — or measure differences the way people see them.

A colour tolerance

A panel is accepted if its difference from the master is small:

$$\Delta E^*_{ab} = \sqrt{(\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2})}$$

People start to notice a difference of about 1–2.

Measuring colour in quality control

A camera records colour; measuring it takes more

Colorimeter and spectrophotometer

Instruments that measure the colour of one spot under a defined light, in CIELAB. Accurate and traceable — the reference for colour tolerances in paint, plastics, textiles, printing.

Colour camera

Every point at once, at line speed — but its RGB values depend on its filters, the light and the white balance. To *measure* colour with it, it must be calibrated.

Making a camera measure colour

- a fixed, specified light — often standard daylight (D65)
- fixed exposure and **white balance**: a camera has no colour constancy (lecture 7)
- a **reference target** of known colours in the field, to calibrate and to check every image
- tolerances in CIELAB, set against the instrument

Examples: sorting LEDs into colour bins · checking the shade of painted parts against a master · grading fruit by ripeness · verifying wire colours in a harness.

PART 4

Files and records

File formats · Compression · Images as quality records

File formats

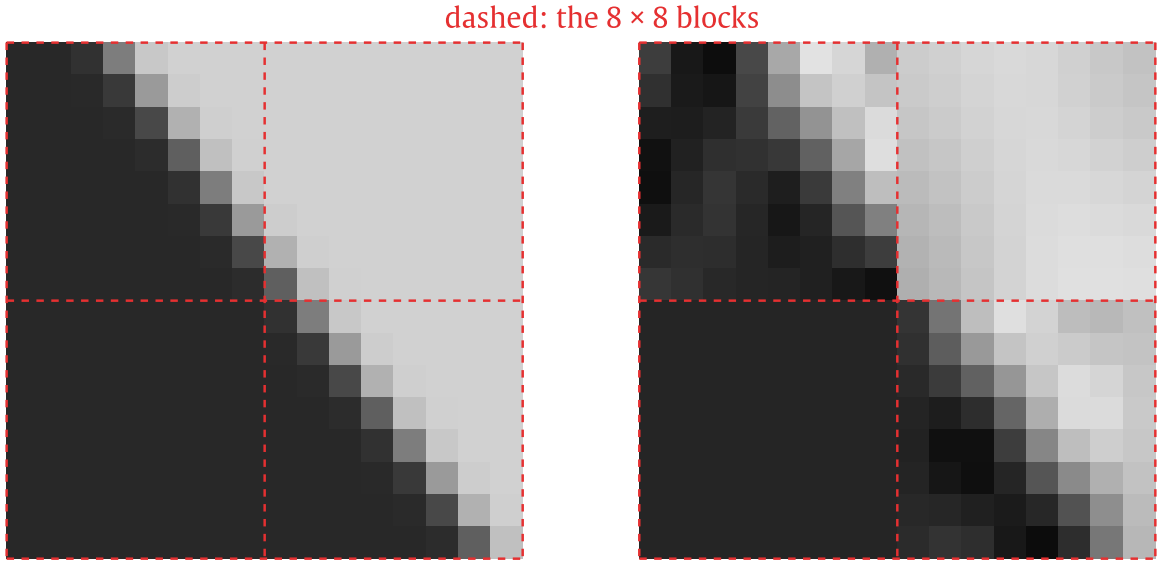
The file decides whether the numbers survive

Format	Compression	Bits per channel	Notes	For measurement?
RAW	none, or lossless	12–14	the sensor's own values, before any processing; camera-specific	**yes**
TIFF	none or lossless	8 or 16	the standard for industry and science; holds metadata	**yes**
PNG	lossless	8 or 16	widespread, compact	**yes**
BMP	none	8	simple, large	yes
JPEG	**lossy**	8	small files; fine detail thrown away	**no**
HEIC	lossy	8–10	phones' default	no
DICOM	lossless or lossy	up to 16	medical images, with patient and device data	with care
MP4 video	lossy, across frames	8–10	recording processes, not measuring them	no

Lossless compression packs the data like a ZIP file: unpacked, every value is back. **Lossy** compression throws away what the eye notices least — which is not what a measurement needs least.

What lossy compression does to an edge

An edge, before and after strong JPEG compression



original **after strong JPEG**

- JPEG cuts the image into blocks of 8 × 8 pixels, describes each block as a sum of stripe patterns, and keeps the coarse ones while rounding away the fine ones.
- Fine patterns are exactly what an edge is made of: after compression the edge has **ripples** beside it and **block borders** across it. An edge found in this image is in a slightly different place.
- The damage grows with the compression and is different in every image: a **random, uncontrolled** error in the measurement.

HOW MUCH DATA

One 8-bit image of 2448 × 2048 pixels: **5.0 MB**; at 30 images per second, **150 MB/s**. Lines read only a region of interest and keep the images that matter — losslessly.

Images as quality records

An image of a rejected part is evidence — if it can be trusted

- **Metadata** travels with the image: time, camera, exposure, gain, lens (the EXIF fields in a photo). Inspection systems add the line, the station, the program version and the unit's **serial number**.
- **What to keep:** images of every rejected unit; a sample of accepted ones; the reference images used to validate the system; for some products, every unit.
- **How long:** as the quality system and the customer require — in medical and automotive supply often for many years.

Data integrity

Regulated industries ask that records be **attributable**, **legible**, **contemporaneous**, **original** and **accurate** — the ALCOA principles of pharmaceutical good manufacturing practice. For images: lossless originals, never edited, with who, when and how attached.

The link to ISO 9001

Documented information that proves conformity must be retained and protected from unintended change (§7.5). A stored image of an inspection is such a record.

Recap TAKEAWAYS

Digital images

1. **Why digital:** a digital image is data — computed on, calibrated, stored with its unit's serial number. It is what makes 100 % end-of-line inspection possible.
2. **Pixels:** each averages the light over its area. Two pixels per period, or detail aliases; 2–3 pixels across a defect. A measurement can still be finer than a pixel.
3. **Grey levels:** 2^b levels, rounding error $\Delta/\sqrt{12}$. Saturation destroys information.
4. **Colour:** three grey images, because we have three cones. Change colour space to separate colour from light; measure colour with a calibrated set-up and CIELAB tolerances.
5. **Files:** lossless for anything measured or kept as a record; JPEG never. An image kept as evidence is an original, with its metadata.

Next
The sensor: how photons become these numbers, and where its noise comes from.

The hands-on lab
Your phone's photos as numbers: channels, grey levels, JPEG and noise.

Questions?

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github.com/LolloCappo/student-hub